

TOTMA, Russian Federation

WMO: 270510

Lat: 59.9806N

Lon: 42.7694E

Elev: 444

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-16.2	-27.8	1.2	-22.7	-20.8	1.8	-16.2	20.8	20.2	17.9	18.3	3.2	10	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	82.7	67.5	79.2	65.8	76.0	64.0	70.4	79.5	67.8	76.0	65.7	73.3	6.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	101.1	75.2	64.7	93.1	72.2	62.6	86.3	69.9	34.6	79.2	32.4	76.1	30.7	73.3	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	15.7	13.5	-27.2	87.6	6.9	3.1	-32.2	89.9	-36.2	91.7	-40.1	93.4	-45.0	95.7		
(5)			-27.1	73.0	6.9	2.7	-32.0	74.9	-36.1	76.5	-40.0	78.1	-45.0	80.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	11.4	14.0	24.1	38.0	50.5	58.2	63.9	58.9	49.3	37.7	25.6	17.6
	DBStd	20.49	14.64	13.69	9.33	8.40	9.20	7.77	6.61	7.00	6.84	7.57	10.59	13.76
	HDD50	5734	1196	1009	804	371	110	16	1	8	93	388	733	1005
	HDD65	10143	1661	1429	1269	811	455	231	101	215	471	847	1183	1470
	CDD50	1191	0	0	0	10	126	262	432	283	72	6	0	0
	CDD65	127	0	0	0	0	6	27	68	25	1	0	0	0
	CDH74	1147	0	0	0	2	83	278	574	206	4	0	0	0
	CDH80	247	0	0	0	0	12	63	121	51	0	0	0	0
	WSAvg	6.1	5.8	6.2	6.6	6.3	6.8	6.1	5.1	5.3	5.7	6.4	6.3	6.4
	PrecAvg	24.3	1.6	1.1	1.2	1.6	1.8	2.8	3.0	2.8	2.4	2.3	1.8	1.8
(15) Precipitation	PrecMax	33.8	3.0	2.1	3.4	3.7	3.5	7.7	5.7	7.2	6.7	4.8	3.4	3.7
	PrecMin	16.8	0.4	0.0	0.1	0.0	0.3	0.5	0.4	0.3	0.9	0.0	0.5	0.6
	PrecStd	3.1	0.6	0.5	0.7	0.9	0.9	1.5	1.2	1.3	1.1	1.1	0.7	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	36.9	48.8	70.5	81.9	86.6	87.8	85.7	73.8	61.7	46.4	41.0
		MCWB	33.6	34.3	40.2	54.7	63.2	69.4	71.2	71.1	62.0	53.5	44.8	40.5
	2%	DB	33.7	34.7	42.4	62.7	75.8	80.9	83.1	79.3	68.0	55.1	41.9	36.9
		MCWB	32.5	32.9	35.7	49.6	60.2	65.3	68.7	66.3	59.0	50.8	40.6	35.6
	5%	DB	32.7	33.2	38.6	56.6	70.9	76.7	80.1	74.9	63.6	51.1	39.6	34.5
		MCWB	31.7	31.8	34.3	46.1	56.9	63.1	67.2	64.4	56.9	47.9	38.5	33.3
	10%	DB	30.5	31.4	36.1	51.9	66.6	72.9	76.9	70.9	60.3	48.2	36.8	32.9
		MCWB	29.3	30.2	32.8	43.7	54.5	61.5	65.5	62.7	55.3	46.1	35.8	31.9
	0.4%	WB	34.2	35.0	41.4	55.5	66.1	70.8	73.8	73.8	63.8	54.7	45.4	40.2
		MCDB	34.5	36.1	46.8	69.2	75.9	82.7	83.0	84.7	70.7	58.6	46.2	40.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.8	33.3	37.1	51.0	61.9	67.7	71.4	68.5	60.3	51.3	40.8	36.0
		MCDB	33.5	34.4	40.4	59.5	72.7	77.6	80.8	76.0	65.6	54.7	41.7	36.9
	5%	WB	31.7	32.0	35.1	47.5	58.9	65.1	68.8	65.7	58.1	48.5	38.4	33.5
		MCDB	32.7	33.1	37.8	55.2	67.9	73.9	76.9	72.1	62.5	50.7	39.3	34.2
	10%	WB	29.3	30.1	33.2	44.3	55.9	62.7	66.8	63.7	55.6	46.0	35.9	32.0
		MCDB	30.4	31.4	35.5	51.1	64.7	70.7	74.6	69.7	59.6	48.2	36.7	32.8
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	12.4	14.8	16.9	18.7	17.7	18.0	16.1	13.4	8.2	6.8	8.5
		MCDBR	8.7	8.5	16.8	25.7	25.5	23.6	23.0	22.1	20.4	14.0	8.1	6.7
	5% WB	MCWBR	8.6	8.0	11.6	15.2	12.6	11.0	10.6	10.7	11.9	9.7	7.5	6.5
		MCDBR	8.3	8.5	12.3	23.4	22.6	21.0	20.6	18.8	17.9	12.8	7.8	6.6
		MCWBR	8.5	8.2	9.0	14.9	12.2	10.8	11.0	10.4	11.2	9.9	7.5	6.5
(40) Clear-Sky Solar Irradiance	taub		0.262	0.304	0.333	0.376	0.360	0.356	0.391	0.380	0.347	0.327	0.286	0.240
	taud		2.043	2.064	2.077	2.183	2.421	2.454	2.368	2.378	2.457	2.374	2.148	2.001
	Ebn at Noon		154	211	246	259	275	278	263	257	247	210	145	113
	Edh at Noon		15	26	35	38	33	32	34	31	25	19	13	10
(44) All-Sky Solar Radiation	RadAvg		104	313	722	1168	1588	1664	1681	1223	697	289	95	54
	RadStd		20	83	116	142	124	155	147	104	83	49	15	10

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.10	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page